

ers train. Apart from measuring and monitoring dozens of game-related metrics, it automatically creates a complete overview of all the activity taking place, right from the delivery of the ball to the bat and up to the final destination after being hit. The data can be viewed on a laptop or phone, or stored online for later analysis.

ProVantage by Loughborough University is such a tool for golfers. It provides golf swing analysis and coaching. The solution uses a 16-camera Vicon motion analysis system, which automatically tracks strategically-placed reflective markers on the golfer's body and club. The swing can be reviewed interactively from any angle and at any speed. Weight distribution—fundamental to golfing—is also monitored using Kistler force platforms embedded on the floor. ProVantage also uses either Tekscan in-shoe sensors or an RSscan mat to provide a more detailed view of pressure distribution beneath each foot. Using such technologies, the solution monitors more than five essential focus areas of golfing including wrist movement and posture.

The Scottish rugby team's coach has gone one step ahead and deployed remotely-operated unmanned aerial vehicles (UAVs) in the form of a radio-controlled helicopter, fitted with high-definition video, high-resolution still cameras, thermal imaging cameras and emission sensors. Formulated by Cyberhawk Innovations, the solution, also called Cyberhawk, helps the coach assess important details using live and recorded relays.

The Cyberhawk can hover beside players, giving their view of the pitch—which helps analyse the positions and patterns of the team. Thermal imaging is used to show the energy levels of players and where and how it gets expended the most. The Cyberhawk uses GPS satellite signals and can fly up to 60 metres in the air. There is a great focus on user control and customisation of the views.

Another research team in Europe

Sensor networks could also be an umpire's aid, but more promising is their application in performance assessment and training, especially of a team as a whole

recently experimented with a chip-fitted football and associated software that tracks and analyses its movement. Such information is also valuable in strategising and training.

Innumerable tools are available for various sports, including tennis, football, cycling and even those like javelin throw. Simple or complex, they all help at some level. Have you played any of those common tennis, cricket or football games on the computer? Even those are a starting point for training as they improve hand-eye coordination.

The third umpire's best friend

In the game of cricket, when on-field umpires are confused, a third umpire's opinion is sought. The third umpire sits off the field in a tech-fortified area, where he can see replays and other information such as the trajectory of the ball, collisions, boundaries and so on, from various angles and at various speeds. Roles similar to the third umpire are permitted in other games too and they also rely on technology for making the critical decisions.

As of now, third umpires mostly rely on replays from various angles to make their decision but other promising technologies are available and have been tried in certain matches, though not yet in mainstream.

Hawk-Eye, for instance, is a complex computer system for predicting the trajectory of a ball. Usually, an umpire is able to see the ball until it hits the batsman's leg, but is not sure where it would have gone thereafter. Hawk-eye could help project the path of the ball 'through' the batsman's legs, and see whether it would have hit the

stumps. This information would be helpful in making 'leg before wicket' decisions.

However, there is a lot of controversy involving the use of such predictions, and hence Hawk-Eye broadcasts are now confined to television replays and not used by umpires. The technology has also been tested successfully for tennis and snooker games and some associations have even accepted it as an umpire's tool.

The Snickometer is a much older and time-tested tool that is very often used by umpires. It uses sound analysis to decide whether a ball has hit the bat or not. A very powerful microphone is fitted in one of the stumps which keeps recording the ambient sounds. The sound of a leather ball striking a willow bat causes a certain sound which can be identified by the Snickometer during a replay.

A much newer technology, again for determining whether a ball has hit a bat or batsman's leg pad, is the Hot Spot infrared imaging system. Two infrared cameras located on the opposite sides of the ground above the field of play continuously record an image. The infrared image shows a bright spot where the ball hits because the friction would have increased the temperature in that area. The technology can increase the effectiveness of replays, especially for umpiring purposes.

Similar technologies are used in various sports. In archery, for example, miniature Doppler radars are embedded in the bows to measure speed and other parameters that could help in training as well as scoring decisions.

Wireless sensor networks promise even more

Speaking of on-field technologies, we cannot ignore the potential of wireless sensor networks to revolutionise cricket, football or any game for that matter.

Many prototypes of wireless sensor networks, especially body sensor networks for sports purposes, have been demonstrated in the recent past.